Fiscal Year:	FY 2005	**Task Last Updated:**	FY 08/30/2006
PI Name:	Snell, Edward H. Ph.D.		
Project Title:	Macromolecule Nucleation and Growth Rate Dispersion Studies:		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Bio-nanotechnology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	esnell@hwi.buffalo.edu	**Fax:**	FY
PI Organization Type:	NON-PROFIT	**Phone:**	716-898-8631
Organization Name:	Hauptman-Woodward Medical Research Institute		
PI Address 1:	Dept. of Structural Biology		
PI Address 2:	700 Ellicott Street		
PI Web Page:			
City:	Buffalo	**State:**	NY
Zip Code:	14203-1102	**Congressional District:**	26
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	00-HEDS-UNSOL
Start Date:	04/01/1999	**End Date:**	05/01/2005
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Miller, Teresa	**Contact Phone:**	256-544-7815
Contact Email:	Teresa.Y.Miller@NASA.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	Dr. Snell the previous P.I. has moved to a new institution and Dr. Mark van der Woerd replaces him as the current P.I. Dr. Snell will still be involved in a subset of the original work.		
COI Name (Institution):	van der Woerd, Mark (BAE Systems Analytical Solutions)		
Grant/Contract No.:	NAG8-1916		
Performance Goal No.:			
Performance Goal Text:			
	The experimental work focuses on nucleation, crystal growth characteristics, and the assessment of x-ray crystal quality for a range of macromolecules. It involves studying the effect of solution conditions on crystal nucleation in order to optimize the number and volume of crystals for x-ray analysis. As part of this work experiments will be performed in microgravity to evaluate the effect of microgravity on the nucleation process. Of particular interest is the relationship between the observed variation in crystal quality and the growth characteristics of the crystals. Ground based crystal growth characteristics will be compared with crystal growth in microgravity using Delta-L. By correlating the crystal growth characteristics in the two environments with the resulting x-ray crystal quality, we expect to recommend ground-based techniques for screening materials which will most benefit from microgravity crystallization. The purpose of this study is to investigate how the formation of macromolecule crystals in solution and their growth characteristics		

Task Description:	effect crystal number, volume, and more importantly, x-ray diffraction quality. Based on the effect of microgravity on these processes it is intended to recommend a screening technique to determine which samples will likely benefit the most from crystallization in microgravity. Knowledge of the molecular structure of macromolecules such as proteins, RNA, DNA, and viruses, leads directly to applications in drug design and the enhanced performance of industrial biological process. From previous research it is becoming clear that microgravity crystallization can provide improved molecular structre information for some macromolecules and makes little difference in others. The development of ground based screening techniques to determine which materials will benefit the most from crystallization in microgravity will allow the optimization of resources and increase the success of macromolecule crystallization experiments on the Space Shuttle Orbiter and International Space Station.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Two papers are now in press from the research as of August 2006. A Quasi-Laue Neutron Crystallographic Study of D-Xylose Isomerase Xylose Isomerase mechanism Meilleur, F., Snell, E.H., van der Woerd, M., Judge, R.A.and Myles, D.A.A. in press European Journal of Biophysics Optimizing crystal volume for neutron diffraction, Snell, E.H., van der Woerd, M.J., Damon, M., Judge, R.A., Myles, D.A.A. and Meilleur, F. in press European Journal of Biophysics. [Ed. note, September 2006: papers now published and listed in Bibliography section.]
Bibliography Type:	Description: (Last Updated: 12/14/2021)
Articles in Peer-reviewed Journals	Meilleur F, Snell EH, van der Woerd MJ, Judge RA, Myles DA. "A quasi-Laue neutron crystallographic study of D-xylose isomerase.erase." European Biophysics Journal. 2006 Sep;35(7):601-9. Epub 2006 May 4. PubMed PMID: 16673077 , Sep-2006
Articles in Peer-reviewed Journals	Snell EH, van der Woerd MJ, Damon M, Judge RA, Myles DA, Meilleur F. "Optimizing crystal volume for neutron diffraction: D-xylose isomerase." European Biophysics Journal. 2006 Sep;35(7):621-32. Epub 2006 May 25. PubMed PMID: 16724195 , Sep-2006